ACADEMIC REPORT

From Return Patterns to Portfolio Performance

Stylized Facts of Asset Returns and Comparative Evaluation of Investment Strategies

PART I — Stylized Facts of Asset Returns

Heavy tails, non-normality and volatility clustering

PART II — Investment Strategy Performance and Benchmarking

Advanced strategies versus equal-weighted and quintile-based strategies, with an S&P 500 ETF buy-and-hold benchmark

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1. Executive Summary and Recommendation

1.1 Scope

This report examines the stylized facts of stock returns and evaluates the performance of alternative portfolio construction approaches, assessing whether more complex investment strategies offer meaningful advantages over simpler strategies and an index-based benchmark. The analysis is organized into two parts.

The first part investigates the empirical behavior of the return series of a selected stock, testing whether well-documented stylized facts—including fat tails, volatility clustering, and asymmetric volatility—are present in the sample.

The second part compares nine portfolio construction strategies, ranging from simple equal-weighted and quintile-based allocations to optimization-based and risk-based approaches. Performance is evaluated using a walk-forward backtest in which portfolio decisions rely exclusively on information available at each decision date. A buy-and-hold investment in an S&P 500 ETF serves as an additional benchmark.

1.2 Key findings

[Summarize the principal stylized facts and the most decision-relevant strategy comparisons. Include only results established by the analysis, with key figures and their measurement period.]

1. Returns for EXXON MOBIL share are not normally distributed, and risk is not constant. Returns for Exxon Mobil are not normally distributed, and risk is not constant. Returns show fat tails, negative skewness at short horizons and highly persistent volatility clustering, and these features persist even at monthly frequency. For the share analysed, daily losses of 5% or more occurred about X times as often as a normal model predicts. These features weaken at monthly frequency. Implication: risk estimates based on historical averages are noisy and change over time, which matters for any strategy that relies on them.

2. Strategy A delivered the [highest median risk-adjusted return / lowest volatility], outperforming the equally weighted benchmark in X% of experiments. However, it required turnover of X% per year, was concentrated in certain periods only.

3. Strategies that rely on expected-return estimates were the most / least stable. Mean–variance and maximum Sharpe ratio portfolios produced the most concentrated allocations and the highest turnover. After costs, they did not / only marginally improve on simpler alternatives.

4. The equally weighted portfolio was a demanding benchmark. Despite using no estimates, it [was beaten consistently only by …].

5. Results depend on the period and the assets. Strategy rankings changed noticeably across experiments, particularly [around market stress / in periods of rising rates]. No strategy dominated in every environment.

If the primary objective is to reduce downside risk while remaining diversified, at low implementation cost, we recommend Strategy X. It offered the best balance between risk reduction, consistency across experiments and turnover. It is not the strategy with the highest historical return, and we do not expect its historical advantage to repeat exactly.

Main caveats: The analysis is historical and uses today's asset universe, which introduces survivorship bias. The 100 experiments overlap in time and are therefore not independent tests. Results also depend on the assumed cost level and rebalancing frequency. These limitations are discussed in Section 6.

1.3 Recommendation to the investment committee

[Recommend an allocation approach, further investigation or no implementation. State the evidence, risks, operational conditions and confidence in the recommendation. Do not assume that an advanced strategy must outperform.]

Investment objective: We assume the goal is a diversified, long-only portfolio with moderate risk, rebalances monthly, and is sensitive to drawdowns and implementation costs. A different objective could lead to a different recommendation.

Recommended strategy: Markowitz Mean Variance Portfolio for the following reasons:

Risk: it delivered [lower volatility and shallower drawdowns] than the equally weighted benchmark in [X]% of experiments.

Consistency: its ranking was [stable] across sampled periods and asset subsets, whereas [Strategy Y]'s results were [highly dispersed].

Implementation: its turnover of [Z]% per year kept transaction costs modest, and its ranking [did not change] under the higher cost assumption.

Robustness to estimation error: it relies only on [risk / correlation] estimates, which are more stable than expected-return estimates.

What we are not claiming. We do not claim that Strategy X will outperform in future, nor that it was the best strategy in every period. Its median Sharpe ratio advantage over the equally weighted benchmark was [modest / within the range of variation across experiments].

Why not the alternatives?

Equally weighted: simple, transparent and a strong benchmark, but it carries higher risk because high-volatility assets receive the same weight as low-volatility ones.

Mean–variance / maximum Sharpe: attractive in theory, but in our tests [their allocations were concentrated and unstable], and [their after-cost performance did not justify the added complexity].

Other strategy: One sentence on result.

2. Data

2.1 Sources, universe and sample

The selected universe of assets comprises a diversified portfolio of USD-denominated and listed securities. Each asset possesses a complete price history from January 2010 onwards unit the end of the observation period, with prices available on every NYSE trading day; consequently, no observations are missing.

The data was extracted on 18 October from Yahoo Finance using the Python package yfinance (version 1.7.0), with the most recent observation used dated 1 October 2026. All calculations employ adjusted closing prices, which adjust historical prices for corporate actions such as stock splits and dividends. Returns computed from adjusted prices therefore represent total returns, under the assumption that dividends are reinvested.

To ensure reproducibility, the raw dataset was downloaded once and stored locally together with the extraction timestamp. All subsequent calculations rely on this stored copy rather than re-downloading the data, as data providers may occasionally revise historical prices.

As noted, the sample retains dates from 1 January 2010 to 1 October 2026 and excludes dates on which no assets are traded. This yields a sample of at least 15 years, as required, and encompasses several distinct market environments, including the European sovereign debt crisis (2011), the COVID-19 market crash (2020), and the inflation and interest-rate shock of 2022.

The selected universe are the following assets via their tickers:

Group	**Tickers**	**Names**
Technology	AAPL, MSFT, ORCL, AMZN, INTC	Apple, Microsoft, Oracle, Amazon, Intel
Healthcare	JNJ, PFE, MRK, UNH, ABT	Johnson & Johnson, Pfizer, Merck, UnitedHealth, Abbott Laboratories
Financials	JPM, BAC, WFC, GS, AXP	JPMorgan, Bank of America, Wells Fargo, Goldman Sachs, American Express
Consumer Staples	PG, KO, PEP, WMT, CL	Procter & Gamble, Coca-Cola, PepsiCo, Walmart, Colgate-Palmolive
Real Assets & Defensive (ETFs & share)	GLD, SLV, TLT, VNQ *(ETFs)*; XOM *(share)*	Gold, Silver, US Treasuries (20+ years), US Real Estate, Exxon Mobil

Tabelle 1-Investment Universe

To conclude, the universe contains 21 individual shares and 4 exchange-traded funds (ETFs). All are listed on US exchanges, trade in US dollars and can be bought directly by an investor.

An ETF holds the underlying securities or assets, trades on an exchange like a share, and delivers the index return minus fees and minus small tracking differences.

Every asset in our universe is investable:

- The 21 shares are held directly, so their returns are what an investor would have earned, apart from transaction costs.
- The 4 ETFs each track a reference that is itself not investable.

- No index is used as an investable asset in this report.

Rationale for the asset selection

The asset universe was chosen to test portfolio construction methods. Portfolio strategies differ mainly in how they respond to differences in risk, return and correlation between assets. A universe of very similar assets would make all strategies look alike, while a universe of illiquid or short-lived assets would make the results difficult to assess and implement. We therefore applied four criteria:

Investability and liquidity: All assets are large, actively traded US-listed shares or ETFs, so the strategies could be implemented at low transaction costs. All assets are liquid, so buying or selling has no price impact.

Data history: As previously stated, every asset has a complete daily data since January 2010 (at least 15 years), so all strategies are evaluated on the same assets over the same periods.

Comparability: All assets trade in US dollars on the same exchange calendar, so no currency or calendar adjustments are needed.

Size: With 25 assets, the backtest in Section 4 can draw many different random subsets of assets, which allows the group to test whether the results depend on the specific assets chosen.

Thus, the group ended up selecting four equity sectors with different economic profiles. The 20 shares are drawn equally from four sectors, five large and established companies each. Two sectors are cyclical such as Technology and Financials and two are stable such as Healthcare and Consumer Staples. The sectors also typically differ in their correlations within, meaning how their members move together.

To add to the previous selection, five assets were further added due to their different response to economic conditions, when comparing with ordinary company shares:

- Gold, as a source of value during market events.
- Silver, an industrial metal.
- US Treasuries, which are considered a safe investment in more volatile times:
- US real estate, an entirely different asset class in this universe which exposes the portfolio to property via REITs.
- Exxon, while being a share representing the energy sector and therefore exposure to oil prices and inflation.

Together, these assets provide the universe with exposure to several sources of risk, including economic growth, interest rates, inflation, and commodity prices.

2.2 Preprocessing of Data

Regarding missing data, the adopted rule specified that, within each asset's history, at most two consecutive missing prices would be forward-filled, generating a zero return on those days. Longer gaps would remain missing and be reported. No backward filling was performed, ensuring that no future information was used.

The resulting clean sample spans 4 January 2010 to 30 September 2026, covering 16.7 years and 4,211 trading days across 25 assets. No asset exhibited filled or remaining missing values; the panel is complete. Accordingly, the missing-data rule did not need to be applied, and no asset has a shorter history than the sample period. The complete return series comprises 4,211 daily, 872 weekly, and 200 monthly observations.

Since adjusted close prices already account for dividends and are corrected for splits, all returns utilized during this analysis are total returns. Moreover, all assets trade in US dollars on US exchanges and follow the NYSE calendar (about 252 trading days per year), so no currency conversion or calendar alignment is required. Returns are expressed from the perspective of a USD-based investor. Weekends and holidays have no observations, so a Friday-to-Monday return spans five calendar days.

Comentado [FLH1]: 3 calender days?

Comentado [GS2R1]: I've corrected for five instead of "three calendar days", cause that's what I believe that the idea was.

2.3 Data quality, Coverage and Survivorship Bias

As described in Section 2.2, all 25 assets have complete data over the full sample, so no assets or observations were excluded.

Cryptoassets were not considered because, in some cases, their price history is shorter than the required 15 years. Moreover, the group considered that cryptoassets such as Bitcoin would introduce abnormally high returns. In addition, no portfolio manager would have recommended investment in this asset class 15 years ago; therefore, its inclusion would be inconsistent with an exercise whose final objective was to identify the most suitable investment strategy for an investment committee.

Unusually high returns were, however, not removed or capped, since large price moves are genuine and central to the analysis performed in the following sections.

In the backtest, a detailed analysis of the presence of different assets across subsets, and of their respective influence on the win rate of one strategy relative to another, showed that each asset appears in at least approximately 33% of cases and usually in no more than 50% of cases. Consequently, no asset is systematically over- or under-represented. All prices were obtained from a single provider and were not cross-checked against a second source. Furthermore, because the universe comprises companies that currently exist, no delisted assets are included, thereby introducing survivorship bias (see Section 6).

Comentado [GS3]: We can talk, on section 6, about how risky strategies could have picked riskier assets that could have not surived, and how that would inevitably influenced returns.

3. Methods and Assumptions

3.1 Part I - Stylized facts of asset returns

3.1.1 Return definitions, frequencies and conventions

Part I examines whether the returns of a single share show the stylised facts widely documented for financial markets. As required by the project brief, we analyse one asset, namely Exxon Mobil (XOM), at daily, weekly and monthly frequency thereby investig ating six features: little or no serial correlation in raw returns, non-normal unconditional distributions with fat tails, asymmetry, including possible negative skewness, volatility clustering, leverage effects, and conditional non-normality once time-varying volatility is accounted for.

Each feature is assessed with charts, descriptive statistics and formal statistical tests. This section sets out the methods and decision rules while the results are reported in Section 4.1, and the formulas and full test output are provided in Appendix B.

The selection of Exxon Mobil was made for the following three reasons. First, it is a large and highly liquid US share with a complete daily price history over the full sample period chosen in our analysis. Thereby, the analysis is not affected by missing data or illiquidity. Second, as the only energy share in our universe, XOM is directly exposed to oil prices, and the sample contains several large commodity shocks: the oil-price collapse of 2014–2016, the COVID-19 crash of 2020, when the front-month WTI futures price briefly turned negative, the energy-price spike of 2022 and not to mention the potent rise in oil prices due to the closing of the strait of Hormuz following the US-Iran war. A share with this exposure is a demanding test case for fat tails, volatility clustering and asymmetric volatility. Third, its return drivers differ from those of the financial, technology, healthcare and consumer shares that make up most of the universe, so it adds information rather than repeating a pattern that other assets already display.

The main limitation of this choice is that the results describe one share. They show how returns behave and which common modelling assumptions are questionable, but they are not a statement about every asset in the universe.

3.1.2 Return definitions, frequencies and conventions

As previously stated, all calculations use adjusted closing prices, so that returns include reinvested dividends and are corrected for stock splits (for more information see section 2.2). For the stylised-fact analysis, continuously compounded (log) returns are utilized:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Weekly and monthly returns are derived from period-end prices (Friday close and the last trading day of the month, respectively). Due to the additivity property of log returns over time, this is identical to summing the daily log returns within each period, which we verified numerically. As previously noted, the dataset comprises approximately 4,211 daily, 872 weekly and 200 monthly observations for the chosen asset. It therefore permits a rich and complete analysis at all three frequencies, which makes it possible to determine whether a feature is specific to short horizons or persists as returns are aggregated.

Three conventions apply throughout. First, skewness and excess kurtosis are computed as sample moment ratios without small-sample bias correction, so that a normal distribution has a value

of zero for both. Second, wherever a chart or test concerns the magnitude of returns (absolute or squared returns), returns are demeaned beforehand, so that the level of returns does not distort the measurement of volatility. Third, volatility is annualised using the square-root-of-time rule (√252, √52 and √12) solely to render the rolling-volatility charts comparable across frequencies; annualised figures are not used for inference.

Simple returns, $R_t = \frac{P_t}{P_{t-1}} - 1$, are employed in Part I only as a robustness check for skewness, because the logarithmic transformation compresses large gains and stretches large losses and may therefore alter the sign of skewness.

3.1.3 Testing framework and decision rules

Each stylised fact is examined at all three frequencies with three types of evidence: a chart that makes the feature visible, a descriptive statistic that measures its size, and, where a suitable statistical test exists, a formal hypothesis test. Table 2 summarises the diagnostics, their settings and the rule used to classify each feature as supported. The rules were fixed before the output was interpreted, and a feature is reported as not supported, or only partly supported, when the evidence does not meet them. All tests use a 5% significance level, and four principles guide the interpretation.

- Economic and statistical significance are distinguished. **With 4,210 daily observations, very small effects become statistically significant; the 95% band for a single autocorrelation coefficient is only about ±0.03. We therefore report the size of each effect, for example the autocorrelation coefficient, the excess kurtosis or the ratio of volatilities, alongside its p-value.**
- Low power at long horizons is acknowledged. **With 200 monthly observations the same band widens to about ±0.14, and the tests can only detect large effects. A non-rejection at monthly frequency is therefore weak evidence that a feature is absent, not proof of its absence.**
- Tests are matched to the dependence in the data. **Several classical tests (Ljung-Box, the D'Agostino skewness test and Jarque-Bera) assume independent and identically distributed observations. Volatility clustering violates this assumption and tends to make these tests reject too often or understate uncertainty. Where this matters, we add robust alternatives: a block bootstrap that preserves volatility clusters, non-parametric tests that do not rely on normality, and an analysis of returns standardised by a rolling volatility estimate.**
- Evidence is judged jointly. **Because several tests are run for each feature and frequency, conclusions rest on whether the chart, the statistic and the tests point in the same direction, not on a single p-value. No formal multiple-testing correction is applied.**

3.1.4 Diagnostics for each stylised fact

Fact 1 — Little or no serial correlation in raw returns

If prices react quickly to new information and expected returns change only slowly, past returns should contain little information about future returns. We first test this jointly with the Ljung-Box Q-statistic, whose null hypothesis is that the first m autocorrelations of log returns are all zero. We use three lag lengths per frequency, corresponding roughly to one week, two weeks and one month for daily data (5, 10 and 20), one month, one quarter and half a year for weekly data (4, 13 and 26), and one quarter, half a year and one year for monthly data (3, 6 and 12).

We then report the lag-1 autocorrelation ρ_1 and compare it with the approximate 95% white-noise band of ±1.96/√T. Because ρ_1^2 equals the share of return variance explained by the previous return, it expresses the economic size of any dependence: a coefficient can be statistically significant and still be irrelevant for an investment decision if it explains only a negligible fraction of the variance. This distinction matters most at daily frequency, where the large sample makes the test very sensitive and where volatility clustering (Fact 4) causes the Ljung-Box test to reject too often. A rejection that is not accompanied by an economically meaningful autocorrelation is therefore not interpreted as evidence of predictable returns. Fact 1 is assessed with statistics and tests only; the autocorrelation function is plotted for absolute returns under Fact 4, where it is the main diagnostic.

Fact 2 — Non-normal unconditional distribution and fat tails

We compare the empirical distribution of returns with a normal distribution that has the same mean and standard deviation. Visually, we first use histograms with the fitted normal density, in which fat tails show up as more mass in the centre and in the tails than the normal curve allows, and then normal QQ plots, in which fat tails appear as points bending away from the reference line at both ends. Numerically, we report skewness and excess kurtosis and apply the Jarque-Bera test, whose null hypothesis is that both are zero, as they are under normality. To translate tail thickness into an intuitive measure, we count how many standardised returns lie more than three standard deviations below and above the mean and compare these counts with the number a normal distribution would produce, about 0.13% of observations per side.

The brief expects fat tails to be most pronounced at daily frequency. Because all statistics are computed identically at the three frequencies, comparing them also shows whether the distribution moves towards normality as returns are aggregated (aggregational Gaussianity). This analysis is unconditional: it describes the distribution of returns over the whole sample without accounting for changes in volatility. Whether fat tails persist once volatility is taken into account is examined separately under Fact 6.

Fact 3 — Asymmetry and negative skewness

Negative skewness means that large losses are more frequent or more extreme than gains of the same size. The sample skewness is the starting point, but it is highly sensitive to a few extreme observations. The standard D'Agostino test assumes normal and independent data; under fat tails and volatility clustering its p-values are far too small. We therefore report it for reference only and base inference on a moving-block bootstrap. The bootstrap resamples blocks of consecutive returns rather than single observations, which keeps volatility clusters intact, and recomputes skewness 2,000 times; the 2.5th and 97.5th percentiles of these estimates form a 95% confidence interval. The block length follows the rule of thumb b ≈ n^(1/3), giving 16, 10 and 6 observations at daily, weekly and monthly frequency, and the random seed (42) is recorded so that the intervals can be reproduced.

Two quantile-based measures complement the moment-based estimate: Bowley skewness, based on the quartiles, and Kelly skewness, based on the 10th and 90th percentiles. Both are unaffected by the most extreme returns and show whether any asymmetry is a feature of the bulk of the distribution or only of a handful of days. Finally, we compare the skewness of log and simple returns. Negative skewness is classified as supported only if the bootstrap interval lies entirely below zero and the robust measures agree in sign.

Fact 4 — Volatility clustering

Volatility clustering means that large returns of either sign tend to be followed by large returns, and calm periods by calm periods. It is therefore assessed through the size of returns rather than their sign. Visually, we plot rolling annualised volatility over windows of 21 trading days, 13 weeks and 12 months (approximately one month, one quarter and one year), and the ACF of absolute demeaned returns up to 60 days, 26 weeks and 24 months. Clustering appears as persistent high- and low-volatility regimes in the first chart and as positive, slowly decaying autocorrelations in the second.

Formally, we apply the Ljung-Box test to squared and to absolute returns at lags of 10 days, 13 weeks and 6 months, and Engle's ARCH-LM test. The ARCH-LM test regresses squared returns on their own past values and tests whether these lags have any explanatory power; its null hypothesis is that there are no ARCH effects. It uses 10, 5 and 3 lags, roughly two weeks, one month and one quarter. Volatility clustering is classified as supported if the autocorrelations of absolute returns are positive and persistent and both tests reject.

Fact 5 — Leverage effect and asymmetric volatility

The leverage effect describes a negative relationship between today's return and future volatility: volatility tends to rise more after losses than after gains of the same size. The classic explanation is that a falling share price raises the firm's financial leverage and therefore the riskiness of its equity. We examine the effect with three complementary tools.

- **Cross-correlations.** We compute the correlation between r_t and future absolute and squared returns, $|r_{t+k}|$ and r^2_{t+k}, for k = 1 to 20 days, 12 weeks and 12 months. Significantly negative values at short lags, outside the $\pm 1.96/\sqrt{T}$ band, indicate a leverage effect.
- **Volatility after down and up moves.** We split the sample by the sign of the previous return and compare the average absolute return after negative and after positive returns. The ratio of the two measures the economic size of the asymmetry. A Welch t-test (two-sided, allowing for unequal variances) and a one-sided Mann-Whitney U test assess its significance; the latter does not rely on normality and is therefore robust to fat tails.

The two tools complement each other: the cross-correlations show how the effect evolves over several horizons, while the comparison of volatility after down and up moves measures its size directly. The feature is classified as clearly supported if the short-lag correlations are significantly negative and volatility is significantly higher after down moves, and as only partly supported if the evidence is significant in one tool or at one frequency only.

Fact 6 — Conditional non-normality

The last fact asks whether returns remain non-normal once their time-varying volatility is taken into account, that is, whether the fat tails found under Fact 2 are fully explained by volatility clustering. The brief asks for this to be assessed with standardised residuals from a suitable volatility model. In line with the scope of the course, we do not estimate GARCH-type models and use a rolling historical-volatility model instead. It is the simplest model consistent with the time-varying volatility documented under Fact 4, and it is widely used in practice.

For each trading day t, the model estimates the conditional mean m_t and conditional volatility s_t as the mean and standard deviation of the previous 60 trading days (approximately one quarter) and standardises the return as $z_t = (r_t - m_t) / s_t$. Because the window ends on day $t - 1$, s_t is a genuine one-day-ahead volatility forecast, and the residuals use no future information. If volatility clustering were the only source of fat tails and returns were conditionally normal, the standardised residuals would be

approximately standard normal, and a residual more than four standard deviations from zero would occur with a probability of only about 0.006% per day, that is, well under once in the whole sample.

We therefore count the residuals with $|z_t| > 4$, compare this count with the number expected under normality, mark the exceedances on the series of daily returns, and list the largest ones with their dates so that they can be linked to market events. Conditional non-normality is classified as supported if exceedances are clearly more frequent than a normal distribution implies, that is, if large shocks remain far more common than normality suggests even relative to the volatility prevailing at the time.

This design has limitations that we state explicitly. It is applied at daily frequency only, because a 60-period window would span more than a year of weekly data and five years of monthly data and would no longer describe current conditions. It relies on an exceedance count rather than a formal normality test of the residuals. Finally, a rolling window adapts to changes in volatility with a delay, so some exceedances at the start of turbulent periods may reflect the lag of the model rather than the true shape of the return distribution.

3.1.5 Assumptions and limitations of the approach

The design rests on several assumptions that should be kept in mind when reading the results in Section 4.1.

- **One share, one history.** All results describe XOM over 2010–2026. A single historical path contains a few dominant episodes, notably 2014–2016 and 2020, which can drive both the tail statistics and the apparent persistence of volatility. The findings illustrate how returns behave but cannot be generalised to the whole universe without further evidence.
- **Constant mean.** Expected returns are treated as constant. This is standard at short horizons, where the mean is small relative to volatility, but slow changes in expected returns are not modelled.
- **Limited data at long horizons.** With 872 weekly and 200 monthly observations, the tests have less power and the GARCH parameters are estimated imprecisely, particularly at monthly frequency. Monthly results are therefore interpreted with caution.
- **Choice of settings.** Lag lengths, window sizes, the outlier threshold and the bootstrap block length follow common conventions and were fixed in advance rather than optimised. Other reasonable choices could change individual p-values, which is why conclusions rest on the overall pattern of evidence.
- **Return definition.** Skewness in particular depends on whether log or simple returns are used, which is why both are reported.
- **Model specification.** The volatility models are standard but deliberately simple. Standardised residuals inherit any misspecification, so remaining non-normality may partly reflect the limits of the model rather than the true distribution of the innovations.

The implications of these properties for portfolio construction, in particular for the choice of estimation windows, risk measures and the reliability of estimated inputs, are discussed at the end of Section 4.1.

[Describe the planned descriptive statistics, distributional diagnostics, normality and tail analysis, return autocorrelation, and volatility-clustering analysis. Define the tests, horizons and significance criteria; move formulas and full diagnostic output to Appendix B.]

We analyse Exxon Mobil (XOM) at daily, weekly and monthly frequency, combining charts, descriptive statistics and formal tests at the 5% significance level (detailed output in Appendix A). As an energy

company, XOM is exposed to oil-price shocks, several of which fall within the sample (the 2014–2016 oil price collapse, the COVID-19 pandemic of 2020 and the 2022 energy prices).
Serial correlation: returns show little linear predictability, with all autocorrelations below 0.11 in absolute value. The daily Ljung-Box test rejects the null of no autocorrelation ($p < 0.001$), but this reflects volatility clustering: on GARCH-standardised residuals the test clearly does not reject ($p = 0.99$). At weekly and monthly frequency no autocorrelation is detected.
Fat tails: returns are fat-tailed at all three frequencies (excess kurtosis 6.65 daily, 4.41 weekly and 2.78 monthly; Jarque-Bera $p < 0.001$ in each case). Daily losses of 5% or more occurred 9.4 times as often as a normal model predicts. Unlike the pattern typically described in the literature, the tails remain heavy at monthly frequency. This is not driven solely by the COVID-19 crash: excluding 2020, monthly excess kurtosis is still 1.67 and normality is still rejected.
Asymmetry: skewness is negative and significant at daily (−0.21) and weekly (−0.40) frequency, but not significant at monthly frequency.
Volatility clustering: this is the strongest feature in the data. The autocorrelation of absolute daily returns is 0.24 at lag 1 and still 0.24 at lag 10, indicating very persistent volatility. The Ljung-Box test on squared returns and the ARCH-LM test reject at all three frequencies.
Leverage effect: the evidence is weak. A GJR-GARCH(1,1) model finds a small but significant asymmetry term at daily frequency ($\gamma = 0.04$, $p < 0.001$), but not at weekly or monthly frequency. After the largest daily falls, next-day volatility is only 12% higher than after rises of equal size. A plausible explanation is that XOM's volatility is driven by oil-price shocks in both directions, not mainly by falls in its own share price.
Conditional non-normality: after standardising returns by GARCH volatility, excess kurtosis falls to 1.40 daily, but normality is still rejected at all three frequencies. A Student-t distribution (estimated $\nu \approx 7$) describes the tails better.
In summary, five of the six stylised facts are clearly supported for XOM, while the leverage effect is weak and only present at daily frequency. In contrast to the common finding that returns become approximately normal over longer horizons, fat tails and volatility clustering persist at monthly frequency, reflecting the commodity-driven nature of the share. For portfolio construction this means that risk is time-varying and extreme losses are more frequent than normal-based models assume, even when portfolios are evaluated monthly.

3.2 Part II - Strategy definitions

[Define each advanced strategy and its economic rationale. Specify inputs, signals, portfolio formation, weighting, constraints, rebalancing, holding periods and execution timing. Define equal weighting and the quintile-based strategy precisely, including ranking variable and whether it is long-only or long–short.]

Notation and assumptions

We consider a universe of N assets. A portfolio is described by its weight vector $w = (w_1, \ldots, w_N)$, where w_i is the share of capital invested in asset i. All inputs are estimated from historical returns within the estimation window of each experiment.

The expected return and the variance of a portfolio are:

$\mu p = w^T \mu = \Sigma_i \, w_i \, \mu_i \; \sigma^2 p = w^T \Sigma w = \Sigma_i \, \Sigma_j \, w_i \, w_j \, \sigma_{ij}$

where:

- μ_i is the expected return of asset i, and μ the vector of all expected returns;
- σ_i is the volatility (standard deviation of returns) of asset i;

- σ_{ij} is the covariance between assets i and j, with $\sigma_{ij} = \rho_{ij}\,\sigma_i\,\sigma_j$, where ρ_{ij} is their correlation (between −1 and +1);
- Σ is the N × N covariance matrix, with the variances σ_i^2 on its diagonal and the covariances σ_{ij} elsewhere;
- C is the N × N correlation matrix, with entries ρ_{ij} and ones on its diagonal;
- $\sigma_p = \sqrt{(w^T \Sigma w)}$ is the portfolio volatility.

To ensure a fair comparison, every strategy is subject to the same two constraints:

$\Sigma_i\, w_i = 1$ (fully invested) $w_i \geq 0$ for all i (long-only, no short-selling)

No leverage is used. The strategies therefore differ only in how they choose the weights, not in what they are allowed to hold. Because each experiment draws a random subset of 10 assets, the equally weighted portfolio shows how much of each asset class is available on average.

Portfolio strategies

a) Equally weighted (EWP): $w_i = 1/N$.

Every asset receives the same capital. The strategy needs no estimates, so it is immune to estimation error, and it serves as our benchmark. Its weakness is that volatile assets dominate portfolio risk.

b) Quintile portfolio (QP): $w_i = 1/K$ for the $K = N/5$ assets with the highest mean past return; $w_i = 0$ otherwise.

At each rebalancing date, assets are ranked by their average daily return over the estimation window, and the top 20% are held in equal weights. We use the long-only version, consistent with the other strategies. The slides also mention ranking by risk-adjusted return and a long–short version, which we do not use.

c) Mean–variance (MVP): $\max\, w^T\mu - (\phi/2)\cdot w^T\Sigma w$, where ϕ is the investor's risk aversion.

The portfolio trades off expected return against variance. We use a risk-aversion parameter set $\phi = 1$, a low penalty on risk. This keeps the portfolio distinct from the next two strategies: the minimum-variance portfolio is the limiting case $\phi \to \infty$, and the maximum Sharpe ratio portfolio corresponds to a higher implied risk aversion. With $\phi = 1$, the MVP is therefore the most return-seeking strategy.

d) Global minimum variance (GMVP): $\min\, w^T\Sigma w$.

The portfolio with the lowest possible risk, ignoring expected returns. It tends to concentrate in low-volatility assets.

e) Maximum Sharpe ratio (MSRP): $\max\, (w^T\mu - r_f) / \sigma_p$, with $r_f = 0\%$.

The portfolio with the highest expected excess return per unit of risk (the tangency portfolio). It is optimal in theory but relies on estimated returns, which makes it prone to concentrated and unstable allocations.

f) Inverse volatility (IVP) and risk parity (ERC):

IVP: $w_i = (1/\sigma_i) / \Sigma_j\, (1/\sigma_j)$

ERC: $w_i(\Sigma w)_i / \sigma p = \sigma p / N$ for all i, where $(\Sigma w)_i$ is the covariance of asset i with the portfolio as a whole.

Both strategies give more weight to less risky assets, but they define risk differently. The inverse-volatility portfolio allocates more capital to assets that are less volatile on their own. The risk parity portfolio allocates capital so that every asset contributes equally to the risk of the portfolio as a whole, which also depends on how assets move together. The two coincide when all correlations are equal, for example when assets are uncorrelated otherwise they generally differ.

g) Most diversified (MDP) and maximum decorrelation (MDCP):

MDP: max $(\Sigma_i w_i \sigma_i) / \sigma_p$

MDCP: min $w^T C w$

The MDP maximises the diversification ratio: the weighted average volatility of the holdings divided by portfolio volatility. The MDCP minimises variance computed on correlations only, ignoring differences in volatility.

The strategies differ in how much they rely on estimated inputs:

- EWP uses none;
- The Quintile portfolio uses only a ranking of past returns;
- IVP and MDCP use only volatilities or correlations;
- GMVP, ERC and MDP use the full covariance matrix;
- MVP and MSRP also require expected returns, the least reliable input.

We expect stability to decrease as reliance on estimates increases.

3.3 Backtesting assumptions and implementation

[Document training and evaluation splits, walk-forward design where used, signal lagging, parameter selection, transaction costs, slippage, financing, shorting and leverage assumptions. Explain controls against look-ahead bias and data leakage.]

Each of the 100 experiments uses a randomly sampled five-year period and a random subset of [K] assets, drawn with a fixed random seed ([seed]) so that all strategies are evaluated on identical data. Weights are estimated on the first [two years], and the strategies are then evaluated out of sample on the final year. Portfolios are rebalanced quarterly, and weights drift with prices between rebalancing dates. Portfolio returns are computed from simple daily returns. Transaction costs are charged at 15 basis points per unit of turnover. Results are reported before and after costs.

3.4 Performance measures and statistical inference

[Define the selected return, risk and risk-adjusted metrics and their annualization conventions. Specify the risk-free rate, inference procedure and treatment of multiple strategy comparisons. Distinguish statistical significance from economic significance.

All strategies are evaluated out of sample on daily adjusted closing prices, net of transaction costs. The fee is charged on the first day of each rebalance as turnover × fee, with turnover defined FORMULA. Portfolio weights drifts between rebalances. The group has computed the for each of the 100 backtest windows and then averaged across windows, so the reported figures are means of per-window statistics and not statistics of a pooled series.

Risk metrics

a) Annualized volatility:

Total risk is measured by the standard deviation of daily returns, annualized with the square-root-of-time rule:

$$\sigma = s(r) \cdot \sqrt{252}$$

b) Downside deviation:

Only returns below the target (here 0) are treated as risk, following the semi-variance concept of Markowitz (1959):

$$\sigma_D = \sqrt{ (1/n) \Sigma \min(r_t, 0)^2 } \cdot \sqrt{252}$$

c) Drawdown and maximum drawdown:

Path-dependent risk is captured by the drawdown, the percentage loss from the running wealth peak:

$$DD_t = W_t / \max_{s \le t} W_s - 1$$

The maximum drawdown is $MDD = -\min_t DD_t$, the largest peak-to-trough loss in the window. The average drawdown is the mean of DD_t over the days spent below a previous peak, reported as a positive number.

d) Value-at-Risk and Expected Shortfall (95%):

The daily VaR is the loss that is exceeded on 5% of days. The Expected Shortfall (CVaR) is the average return on the days beyond the VaR threshold, so it measures the severity of the tail and not only its location (Rockafellar and Uryasev, 2000). Both are reported as daily figures.

e) Distribution diagnostics:

We also report skewness and excess kurtosis, which describe the asymmetry and tail thickness of the return distribution. These are not risk measures in their own right, but they document the stylized facts of heavy tails and gain/loss asymmetry in asset returns.

Risk-adjusted metrics

Risk-adjusted metrics scale a measure of return by a measure of risk. They are defined as follows, where $\bar{r}$ is the mean daily return and $CAGR = W_T^{(365/days)} - 1$ is the geometric annualized return.

a) Sharp ratio

The **Sharpe ratio** (Sharpe, 1966) is the benchmark measure and treats upside and downside deviations symmetrically, which is denominated by volatility.

$$SR = (\bar{r} - r_f) / s(r) \cdot \sqrt{252}$$

b) Sortino ratio

The **Sortino ratio** corrects this by penalizing only returns below the target, so it favours positively skewed strategies relative to the Sharpe ratio.

$$(\bar{r} - r_f) / \sigma_{D,daily} \cdot \sqrt{252}$$

c) Calmar ratio

The **Calmar** and **Sterling ratios** judge return against drawdown risk, in this case Calmar uses the single worst episode and is therefore more sensitive to the length of the window, therefore calculating the maximum drawdown.

$$CAGR / MDD$$

d) Sterling ratio

The Sterling ratio uses the average drawdown and is more stable in comparison with the Calmar ratio due to be the averge drawdon instead of the maximum drawdown.

$$\text{Average annual return} / \text{average drawdown}$$

The ranking of this strategies in the results is based on average annual return, volatility, Sharpe, Sortino, Calmar and Sterling ratios, and maximum drawdown. VaR, CVaR, skewness, kurtosis and the daily win rate are reported as complementary diagnostics.

Risk-free rate

The risk-free rate is set to zero throughout the study ($r_f = 0$). It is applied consistently in three places: the excess-return numerator of the Sharpe ratio, the target return in the Sortino ratio, and the

objective of the Maximum Sharpe Ratio portfolio, whose optimization therefore maximizes $\mu'w / \sqrt{(w'\Sigma w)}$. The Calmar and Sterling ratios do not use a risk-free rate.

Inference procedure

The unit of analysis and pairing inference is based on the 100 backtest windows. In every window all strategies and the benchmarks (1/N, Quintile, S&P 500 ETF) use the same dates and the same asset subset, so the comparison is paired.

A strategy "wins" in window i when $d_i > 0$ for metrics where higher is better (returns and the Sharpe, Sortino, Calmar and Sterling ratios) and when $d_i < 0$ for volatility and maximum drawdown. The group report the win rate (share of windows won), the mean and median of d_i, and the dispersion of the Sharpe ratio across windows (box plots). The group has also split results by out-of-sample start year and by the presence of individual stocks in the window, to see whether the outcome depends on a specific period or asset.

The 100 windows are random 3-year blocks drawn from the same 15 years, so they overlap heavily and share market regimes. They are not 100 independent observations, and the usual standard error of the win rate (about 5 percentage points) understates the true uncertainty. We therefore build the confidence interval for the mean difference $E[d_i]$ with a cluster bootstrap that resamples windows by out-of-sample start year. A paired Wilcoxon signed-rank test is reported as a complementary check, with the caveat that its p-values remain optimistic under overlap.

Statistical versus economic significance

Statistical significance asks whether an observed difference between a strategy and a benchmark is larger than what sampling noise could produce. It is assessed with the paired, dependence-robust tests and multiplicity adjustments described above, and it depends on the size of the difference, the variability of returns and the (effective) number of independent observations. It says nothing about whether the difference matters to an investor. With a long enough sample a very small difference becomes significant, and with a short, noisy sample like ours (3-year out-of-sample windows, overlapping across a 15-year history) a large difference can fail to be.

Economic significance asks whether the difference is large enough to matter after the costs and constraints an investor actually faces. We judge it on:

- Net-of-cost return,is the excess return over 1/N after transaction costs, in percentage points per year and in terminal wealth over the 3-year horizon.
- Cost sensitivity, is the cost drag (annual turnover × fee) and the break-even fee at which a strategy's advantage disappears.
- Risk, is whether the extra return comes with higher volatility, larger drawdowns or fatter tails (Sharpe, Sortino, Calmar, maximum drawdown), so that a higher mean return is not mistaken for a better outcome.
- Stability, is whether the advantage holds across windows and periods, or is driven by a few outlier windows.

4. Results

4.1 Part I — Evidence on asset-return properties

[Present the main findings on heavy tails, non-normality, volatility clustering and other investigated properties. Use a small set of summary exhibits, explain their economic meaning and link them to the assumptions used in Part II.]

This section reports the results of the diagnostics set out in Section 3.1, following the same order of stylised facts. Each subsection presents the relevant statistics for Exxon Mobil (XOM) at daily, weekly and monthly frequency, and Table 9 at the end of the section summarises whether each feature is supported. Full test output is available in the accompanying notebook.

4.1.1 Fact 1 — Little or no serial correlation in raw returns

Table 3 – Serial correlation in XOM log returns

Frequency	Ljung-Box lags m	Q(m)	p-value	ρ_1	95% band	ρ_1^2 (variance explained)
Daily	5 / 10 / 20	13.1 / 30.2 / 55.8	0.023 / 0.001 / <0.001	−0.020	±0.030	0.04%
Weekly	4 / 13 / 26	0.9 / 20.5 / 39.8	0.921 / 0.083 / 0.041	−0.003	±0.066	0.00%
Monthly	3 / 6 / 12	0.2 / 0.9 / 7.7	0.985 / 0.990 / 0.810	+0.010	±0.139	0.01%

The evidence supports this feature at all three frequencies (Table 3). At daily frequency, the Ljung-Box test rejects the null hypothesis of no autocorrelation at all three lag lengths. The dependence behind these rejections is economically negligible, however. The lag-1 autocorrelation is −0.020, inside the 95% white-noise band of ±0.030, and explains only 0.04% of the variance of daily returns. In line with the decision rule in Section 3.1, the daily rejections are therefore not interpreted as evidence of predictable returns. They are consistent with the very strong volatility clustering documented under Fact 4, which inflates the rejection rate of the Ljung-Box test. At weekly frequency, the test rejects only at the longest lag (26 weeks, $p = 0.041$) and not at 4 or 13 weeks, and the lag-1 autocorrelation is essentially zero. At monthly frequency, no test comes close to rejection ($p \geq 0.81$). For an investor, past XOM returns contain no information that could be used to forecast next-period returns.

4.1.2 Fact 2 — Non-normal unconditional distribution and fat tails

Table 4 – Distribution and tail statistics of XOM log returns

Frequency	n	Skewness	Excess kurtosis	Jarque-Bera (p-value)	Returns below −3σ	Returns above +3σ	Normal expect. per side
Daily	4,210	−0.21	6.64	7,763 (<0.001)	37	25	5.7
Weekly	873	−0.40	4.42	732 (<0.001)	7	8	1.2
Monthly	200	−0.17	2.78	65 (<0.001)	1	2	0.3

Note: skewness and excess kurtosis are sample moment ratios without bias correction.

This feature is clearly supported at all three frequencies and, as expected, is strongest at daily frequency (Table 4). Daily returns have an excess kurtosis of 6.64, and the Jarque-Bera test rejects

normality decisively ($p < 0.001$). The histograms and QQ plots (Figure 1) show the typical pattern of a sharper peak and heavier tails than the fitted normal distribution, with the QQ points bending away from the reference line at both ends. The tail counts make the difference concrete. A normal distribution implies about 11 daily returns beyond three standard deviations over the sample, whereas XOM recorded 62, more than five times as many. The left tail, with 37 such returns, was heavier than the right tail, with 25. The largest single loss was a daily log return of −13.0% on 9 March 2020.

Tail thickness declines as returns are aggregated: excess kurtosis falls from 6.64 at daily to 4.42 at weekly and 2.78 at monthly frequency, the pattern known as aggregational Gaussianity. The convergence is far from complete, however. Normality is still rejected at monthly frequency (JB = 65, $p < 0.001$), and the number of returns beyond three standard deviations remains five to six times the normal expectation at every frequency. With only 200 monthly observations, a few extreme months around the 2020 crash carry considerable weight, so the monthly figures should be read with some caution. Even so, fat tails are not confined to short horizons for this share.

4.1.3 Fact 3 — Asymmetry and negative skewness

Table 5 – Skewness of XOM returns: moment-based, bootstrap and robust measures

Frequency	Skewness (log returns)	D'Agostino p (assumes normality)	Block-bootstrap 95% CI	Bowley skewness	Kelly skewness	Skewness (simple returns)
Daily	−0.21	<0.001	[−0.77; 0.36]	0.03	0.00	−0.00
Weekly	−0.40	<0.001	[−1.08; 0.28]	−0.02	−0.03	−0.08
Monthly	−0.17	0.311	[−0.89; 0.63]	0.04	−0.07	+0.30

Negative skewness is not supported by robust evidence at any frequency (Table 5). The moment-based skewness of log returns is negative at all three frequencies, and the D'Agostino test classifies the daily and weekly values as highly significant. This test assumes normal and independent observations, however, and is therefore unreliable for these data. The moving-block bootstrap, which preserves volatility clusters, gives 95% confidence intervals that include zero at every frequency. The robust quantile-based measures point the same way: Bowley and Kelly skewness are close to zero and change sign across frequencies. The result also depends on the return definition. With simple returns, skewness is essentially zero at daily and weekly frequency and turns positive at monthly frequency (+0.30).

Taken together, the negative point estimates reflect a small number of large losses rather than a systematic asymmetry in the bulk of the distribution. Those large losses are already captured by the fat-tail evidence under Fact 2.

4.1.4 Fact 4 — Volatility clustering

Table 6 – Volatility clustering in XOM returns

Frequency	Lag	Ljung-Box on r^2: p-value	Ljung-Box on $\lvert r \rvert$: p-value	ARCH-LM lags	ARCH-LM statistic (p-value)	ACF of $\lvert r \rvert$, lag 1	ACF lags above 95% band
Daily	10	<0.001	<0.001	10	877.1 (<0.001)	0.25	60 of 60
Weekly	13	<0.001	<0.001	5	117.1 (<0.001)	0.23	18 of 26

Frequency	Lag	Ljung-Box on r^2: p-value	Ljung-Box on $\|r\|$: p-value	ARCH-LM lags	ARCH-LM statistic (p-value)	ACF of $\|r\|$, lag 1	ACF lags above 95% band
Monthly	6	0.002	0.005	3	16.4 (<0.001)	0.24	7 of 24

Volatility clustering is the strongest feature in the data and is supported at all three frequencies (Table 6). The rolling-volatility chart (Figure 2) shows clearly separated regimes. In calm phases, notably 2013 and 2017, the annualised 21-day volatility was mostly between 10% and 15%. In turbulent phases, namely the oil-price collapse of 2014–2016, the COVID-19 crash of 2020 and the energy-price shock of 2022, it rose to several times that level and peaked above 100% in March 2020. The autocorrelation of absolute daily returns (Figure 3) is about 0.25 at lags 1 to 10 and still about 0.13 after 60 trading days, and all 60 lags lie above the 95% band of ±0.030. This slow decay shows that volatility shocks persist for months.

The formal tests confirm the visual evidence. At daily frequency, the Ljung-Box tests on squared and absolute returns and the ARCH-LM test reject the null hypothesis of no clustering, with p-values indistinguishable from zero. The effect weakens with aggregation but remains significant throughout. At weekly frequency, 18 of the 26 autocorrelations of absolute returns lie above the band and all tests reject. At monthly frequency, the lag-1 autocorrelation of absolute returns is still 0.24, and both the Ljung-Box tests ($p = 0.002$ and 0.005) and the ARCH-LM test ($p < 0.001$) reject.

4.1.5 Fact 5 — Leverage effect and asymmetric volatility

Table 7 – Volatility after negative and positive returns, and leverage correlations

Frequency	Mean $\|r\|$ after down move	Mean $\|r\|$ after up move	Ratio down / up	Welch p (two-sided)	Mann-Whitney p (one-sided)	Lag-1 leverage correlation: $\|r\|$ / r^2	95% band
Daily	1.14%	1.06%	1.07	0.027	0.027	−0.04 / −0.06	±0.030
Weekly	2.54%	2.41%	1.06	0.410	0.045	−0.11 / −0.11	±0.066
Monthly	5.10%	4.91%	1.04	0.791	0.473	−0.15 / −0.21	±0.139

The leverage effect is present but economically small, and it is clearly supported only at daily frequency (Table 7). The leverage correlations (Figure 4) are predominantly negative at short horizons. At daily frequency, the correlation between today's return and the absolute return k days later lies between −0.04 and −0.08 for k = 1 to 6, below the lower band of −0.030, and the correlation with squared returns is strongest at lag 2 (−0.10). At weekly and monthly frequency, the correlations are significantly negative mainly at the first lag.

The direct comparison of volatility after down and up moves confirms the direction of the effect but shows that it is modest. After a negative day, the average absolute return is 1.14%, compared with 1.06% after a positive day, a ratio of 1.07, and the difference is significant in both tests ($p = 0.027$). At weekly frequency the ratio is 1.06; it is significant in the one-sided Mann-Whitney test ($p = 0.045$) but not in the Welch test ($p = 0.41$). At monthly frequency the ratio of 1.04 is insignificant in both tests.

Under the decision rule in Section 3.1, the leverage effect is therefore clearly supported at daily frequency and only partly supported at weekly and monthly frequency. A plausible reason for its small

size is that XOM's volatility is driven largely by oil-price news, which moves the share sharply in both directions, as the daily log return of +11.9% on 9 November 2020 illustrates.

4.1.6 Fact 6 — Conditional non-normality

Table 8 – Largest daily returns relative to trailing 60-day volatility (|z| > 4)

Date	Log return	z	Context
2 Feb 2018	−5.2%	−7.33	Broad equity sell-off, early February 2018
31 Jul 2014	−4.3%	−6.26	–
9 Mar 2020	−13.0%	−6.01	Oil-price collapse after breakdown of OPEC+ talks
9 Nov 2020	+11.9%	+6.01	COVID-19 vaccine announcement
5 Feb 2018	−5.9%	−5.87	Equity volatility shock, February 2018
28 Jul 2015	+4.0%	+5.73	–
31 Jul 2015	−4.7%	−5.27	–
8 Aug 2011	−6.4%	−4.83	Sell-off after US sovereign credit-rating downgrade
4 Apr 2025	−7.5%	−4.81	Market turmoil after US tariff announcements
28 Sep 2016	+4.3%	+4.54	OPEC agreement to cut output
All exceedances: 18 observed vs. 0.3 expected under normality (4,150 days; 13 negative, 5 positive)			

Note: $z = (r_t - m_t) / s_t$, with m_t and s_t the mean and standard deviation of the previous 60 trading days. The table shows the ten largest of the 18 exceedances.

Conditional non-normality is supported at daily frequency (Table 8). When each daily return is standardised by the mean and standard deviation of the previous 60 trading days, a normal distribution would produce about 0.3 residuals beyond four standard deviations over the 4,150 days for which a residual can be computed. XOM produced 18 such residuals (Figure 5), roughly 60 times the normal expectation; 13 of them are losses and five are gains. Large shocks therefore remain far more frequent than normality implies, even relative to the volatility prevailing at the time. Volatility clustering accounts for part of the unconditional fat tails found under Fact 2, but not for all of them.

Several of the largest exceedances can be linked to identifiable market events. The largest, at −7.3 standard deviations, occurred on 2 February 2018 and was followed by another on 5 February 2018, both during the broad equity sell-off of early February 2018, which followed an unusually calm year for the share. The log return of −13.0% on 9 March 2020, after the breakdown of the OPEC+ talks, and the log return of +11.9% on 9 November 2020, on news of an effective COVID-19 vaccine, both exceed six standard deviations. Further exceedances fall in the sell-off of August 2011 that followed the downgrade of the US sovereign credit rating, in the market turmoil of early April 2025, and on the day of the OPEC agreement to cut output in September 2016.

These results carry the limitations noted in Section 3.1. The evidence is daily only and rests on an exceedance count rather than a formal test, and the rolling window reacts to changes in volatility with a delay. Some exceedances at the start of turbulent periods, such as those in February 2018, are therefore partly a consequence of the model's lag. Given the size of the gap between 18 observed and 0.3 expected exceedances, however, this lag is unlikely to explain the result.

4.1.7 Summary and implications for Part II

Table 9 – Summary of the evidence on the six stylised facts (XOM)

Stylised fact	Daily	Weekly	Monthly	Overall assessment
1 Little or no serial correlation	Supported (tests reject, but ρ_1 negligible)	Supported	Supported	**Supported**
2 Non-normality, fat tails	Supported (strongest)	Supported	Supported (weaker)	**Supported**; weakens with aggregation but persists
3 Negative skewness	Not supported	Not supported	Not supported	**Not supported**; negative point estimates not robust
4 Volatility clustering	Supported (very strong)	Supported	Supported (weaker)	**Supported**; strongest feature in the data
5 Leverage effect	Supported (small effect)	Partly supported	Partly supported	**Partly supported**; mainly a modest daily effect
6 Conditional non-normality	Supported	–	–	**Supported** at daily frequency

Four of the six stylised facts are clearly supported for XOM, one is only partly supported and one is not supported. Unlike the typical pattern described in the literature, fat tails and volatility clustering remain visible at monthly frequency. Negative skewness, often listed as a standard feature of equity returns, is not robustly present for this commodity-driven share.

These findings have direct consequences for the portfolio analysis in Part II. First, volatility is strongly time-varying and persistent, so risk estimated over a past window can differ markedly from the risk experienced afterwards, especially when markets move from a calm into a turbulent regime. This supports re-estimating portfolio inputs on a rolling basis, as in the walk-forward backtest. It also means that all strategies relying on estimated risk adjust to new regimes with a delay. Second, extreme losses occur far more often than a normal distribution implies, both unconditionally and relative to current volatility, so volatility and the Sharpe ratio alone understate downside risk. This is why Part II also reports maximum drawdown and drawdown-based ratios. Third, returns show no exploitable serial correlation, and their mean is estimated much less precisely than their volatility. Strategies that rely on estimated expected returns, such as the mean-variance and maximum Sharpe ratio portfolios, are therefore more exposed to estimation error than strategies that rely only on risk estimates. These conclusions are drawn from a single share and are not a statement about every asset in the universe.

The five financial stocks are highly correlated with each other (average monthly correlation [0.73]), because they share the same exposure to interest rates and credit conditions. Technology and healthcare stocks are much less correlated (about [0.38] and [0.36]. Correlations between sectors are lower still, for example [0.16] between technology and healthcare. This variation is what diversification-based and risk-based strategies are designed to exploit.

Comentado [LT4]: Update data

Comentado [gu5R4]: Here, we should divide this chapter into 2. The first talking about general properties of our assets. The second will will then be the analysis on the returns of our chosen asset

4.2 Part II — Comparative strategy performance

[Present a consistent net-of-cost comparison of all strategies and benchmarks over the common evaluation sample. Discuss cumulative performance, drawdowns, risk-adjusted results and the drivers of observed differences.]

All nine strategies and the S&P 500 ETF are evaluated on the same 100 out-of-sample windows. Each window has the same dates and the same 10-asset subset for every strategy. Weights are estimated on a 2-year lookback, followed by 1 year of out-of-sample trading with quarterly rebalancing and weights drifting in between. A proportional cost of 15 bps is charged on turnover, including the initial investment. The ETF is held buy-and-hold over the same dates. Figures are averages of per-window metrics.

Tabelle 2 Net-of-cost performance (quarterly rebalancing, 15 bps), sorted by Sharpe ratio

Strategy	Ann. return	Cum. return	Volatility	Max DD	Sharpe	Sortino	Calmar	Turnover (p.a.)	Cost drag (pp p.a.)
1/N	13.07%	43.43%	15.23%	19.62%	0.903	1.289	0.762	0.25	0.09
Risk Parity	10.76%	35.46%	12.54%	16.96%	0.900	1.282	0.751	0.26	0.09
Inverse Volatility	11.96%	39.46%	13.99%	18.35%	0.899	1.281	0.759	0.25	0.09
Max Decorrelation	11.35%	37.55%	13.44%	18.07%	0.884	1.268	0.716	0.56	0.13
S&P 500 ETF (buy & hold)	*14.33%*	*46.81%*	*16.76%*	*21.95%*	*0.880*	*1.241*	*0.770*	–	–
Most Diversified	10.07%	33.06%	12.16%	16.89%	0.875	1.254	0.701	0.54	0.13
GMVP	8.34%	26.40%	11.67%	16.88%	0.765	1.086	0.598	0.59	0.14
Quintile	16.17%	55.05%	22.81%	25.89%	0.717	1.041	0.631	2.07	0.36
Markowitz	17.90%	61.73%	26.21%	28.99%	0.680	0.997	0.620	2.37	0.41
Max Sharpe	9.82%	30.57%	15.90%	19.52%	0.630	0.900	0.532	1.93	0.34

Regarding cumulative performance Markowitz (61.7%) and Quintile (55.1%) deliver the highest 3-year cumulative returns, followed by the ETF (46.8%) and 1/N (43.4%). The risk-based strategies fall between 33% and 39%, and GMVP (26.4%) and Max Sharpe (30.6%) are lowest. The ordering by cumulative return is almost the ordering by volatility, so the extra return of the concentrated strategies is compensation for risk and not clearly a free gain. Costs reduce cumulative returns by only 0.4 pp (1/N) to 1.8 pp (Markowitz) over three years, and they leave the return ranking unchanged across the 0–50 bps range tested.

Also in the drawdowns Markowitz (29.0%) and Quintile (25.9%) have the deepest average maximum drawdowns, deeper than the ETF (22.0%). The diversification-oriented strategies are shallowest: Most Diversified and GMVP (about 16.9%), Risk Parity (17.0%), Max Decorrelation and Inverse Volatility (about 18%). 1/N (19.6%) and Max Sharpe (19.5%) sit in between. Risk Parity beats 1/N on maximum drawdown in 74% of windows and Inverse Volatility in 80%, against 9% for Markowitz and 15% for Quintile.

Now looking at the risk-adjusted results the group has come with the present results:

- The top group is 1/N, Risk Parity and Inverse Volatility are essentially tied (Sharpe 0.903, 0.900, 0.899), with Max Decorrelation, the ETF and Most Diversified close behind (0.875–0.884). The

gaps within this group are far smaller than the dispersion across windows (the Sharpe interquartile range is about 0.5 in the box plots), so we do not read any ranking within it as evidence. No strategy beats 1/N convincingly: the best Sharpe win rate against it is 56% (Max Decorrelation).

- According to the Calmar the ETF has the highest Calmar (0.770), slightly ahead of 1/N (0.762) and Inverse Volatility (0.759).
- The bottom group is GMVP (0.765), Quintile (0.717), Markowitz (0.680) and Max Sharpe (0.630) all fall clearly below 1/N. They beat it on Sharpe in only 35%, 31%, 33% and 17% of windows respectively.
- When rebalancung It improves the Sharpe ratio of the diversified strategies, from 0.886 to 0.903 for 1/N, 0.872 to 0.900 for Risk Parity and 0.877 to 0.899 for Inverse Volatility (comparing the no-rebalancing and 15 bps scenarios). It hurts the return-driven ones: Quintile falls from 0.840 to 0.717, Markowitz from 0.791 to 0.680 and Max Sharpe from 0.805 to 0.630.

4.3 Implications for portfolio allocation

[Explain whether additional complexity is justified after costs, risk, turnover and implementation burden. Discuss exposure differences, diversification and consistency with the committee's investment objective.]

Main comparison table — fill with results

Strategy	Ann. return	Ann. volatility	Sharpe ratio	Max. drawdown	Turnover
[Advanced strategy 1]	[Result]	[Result]	[Result]	[Result]	[Result]
[Advanced strategy 2]	[Result]	[Result]	[Result]	[Result]	[Result]
Equal-weighted	[Result]	[Result]	[Result]	[Result]	[Result]
Quintile-based	[Result]	[Result]	[Result]	[Result]	[Result]
S&P 500 ETF buy-and-hold	[Result]	[Result]	[Result]	[Result]	[Result]

[Caption: state evaluation dates, currency, return and annualization conventions, risk-free rate, cost treatment and turnover definition. Add or remove strategy rows as needed.]

5. Robustness Checks and Limitations

5.1 Robustness checks

[Evaluate sensitivity to sample periods, market conditions, costs, rebalancing frequency, parameter choices and alternative reasonable specifications. Identify checks completed versus checks not performed.]

To test whether our conclusions depend on specific assumptions, we repeated the analysis under alternative settings:

- **Higher transaction costs ([25] bp):** rankings [remained unchanged / changed]. [The return-based optimisers fell from rank x to rank y], whereas the low-turnover strategies were barely affected.
- **Sub-periods:** we compared experiments evaluated before and after 2022. [Risk-based strategies lost much of their advantage after 2022], consistent with the reversal in the stock–bond correlation (from [−0.24] to [+0.30]).
- **Asset subsets:** we compared experiments that include and exclude Treasuries and gold. [The benefit of risk-based strategies was concentrated in subsets containing these assets.]
- **Risk aversion** [optional]: varying ϕ between [2] and [25] [changed / did not change] the mean–variance portfolio's ranking.
- **Rebalancing frequency** [optional]: quarterly instead of monthly rebalancing reduced turnover by [x]% with [little / some] effect on risk.

Overall, conclusions [a, b] proved robust, whereas [c] depended on [the cost assumption / the sample period].

5.2 Limitations and residual risks

[Discuss sample size, dependence, selection bias, multiple testing, model risk, capacity, liquidity and benchmark comparability where relevant. Explain how each material limitation affects confidence in the conclusions.]

The results should be interpreted with the following limitations in mind:

- Historical evidence: backtests describe past conditions. Our analysis of returns (Section 3) shows that volatility and correlations change over time, so estimated inputs may not reflect future conditions.
- Overlapping experiments: the 100 experiments share historical periods and are not independent, which limits the statistical confidence of the comparisons.
- Survivorship and hindsight bias: the universe was selected in 2026 from companies that are successful today. This flatters the returns of all strategies, and possibly most those that concentrate in past winners.
- Universe composition: 84% of assets are shares and there is only one bond ETF. Differences between strategies therefore partly reflect different levels of equity exposure, and results may not generalise to a more balanced or international universe.
- Simplified costs: costs are modelled as a fixed rate per unit of turnover. Market impact, bid–ask spreads in stressed markets, taxes and management fees are not included.
- Estimation choices: results depend on the estimation window, rebalancing frequency, risk-aversion parameter and risk-free rate. These were fixed in advance rather than optimised. Using sample estimates without shrinkage is likely to disadvantage the optimisation-based strategies.
- Long-only constraint: excluding short-selling and leverage makes the strategies implementable, but it also limits how far the optimisations can depart from simple allocations.

6. Conclusions

6.1 Answers to the research questions

[Synthesize what Part I establishes and whether Part II supports the investment case for greater strategy complexity. Avoid introducing new results.]

This report examined the behaviour of financial market returns and compared eight portfolio construction strategies under identical conditions.

The analysis of Exxon Mobil's returns (Section 3) confirmed most of the documented stylised facts. Returns show fat tails, negative skewness at short horizons and strongly persistent volatility clustering. Large daily losses occurred about nine times as often as a normal distribution predicts. Unlike the common finding in the literature, these features persisted at monthly frequency. Risk is therefore neither constant nor normally distributed, and estimates based on historical averages are noisy.

The backtest across [100] experiments (Section 5) showed that [Strategy A] achieved the [highest median risk-adjusted return / most consistent outperformance of the benchmark], beating the equally weighted portfolio in [x]% of experiments. [The risk-based strategies reduced volatility and drawdowns in most experiments], while [the return-based optimisers were the most concentrated, traded most heavily and did not deliver a consistent advantage after costs]. No strategy outperformed in every period or asset subset.

Three conclusions follow for an investor choosing a portfolio construction method:

6.2 Decision and next steps

[Restate the supported decision, conditions for implementation, further validation needed and monitoring criteria. Remain consistent with the executive summary.]

On this basis, for an investor seeking [a diversified, long-only portfolio with moderate risk, sensitive to drawdowns and implementation costs], we recommend [Strategy X]. It offered [the best balance between risk reduction, consistency across experiments and turnover]. It was not necessarily the strategy with the highest historical return, and we do not expect its historical advantage to repeat exactly. The recommendation should be reviewed if the investor's objective changes, if correlations between asset classes shift materially, if costs differ substantially from our assumptions, or if live results move outside the range observed in the backtest.

7. References

[Insert all cited academic papers, data-provider documentation, ETF documentation and software references using one consistent citation style. Include access dates where appropriate.]

Bravo, J. M. (2026a). Primer on financial data and heuristic portfolios (Lecture slides). Asset Pricing & Portfolio Management, NOVA IMS.

Bravo, J. M. (2026b). Markowitz's modern portfolio theory (Lecture slides). Asset Pricing & Portfolio Management, NOVA IMS.

Bravo, J. M. (2026c). Risk-based portfolios (Lecture slides). Asset Pricing & Portfolio Management, NOVA IMS.

8. Reproducibility and Technical Appendix

Appendix A — Data dictionary and provenance

[Provide variable definitions, sources, extraction dates, identifiers, transformations and data-access instructions. Explain any licensing restrictions.]

Appendix B — Detailed methods and diagnostics

[Include formulas, hypotheses, test settings, full statistical output and additional distributional, autocorrelation and volatility diagnostics.]

Appendix C — Strategy specifications and backtest configuration

[Provide full strategy rules, parameter values, optimization settings, constraints, transaction-cost assumptions and the complete strategy/benchmark inventory.]

Appendix D — Supplementary results and robustness output

[Include full performance tables, subperiod results, sensitivity analyses and additional figures. Cross-reference every appendix exhibit used in the main report.]

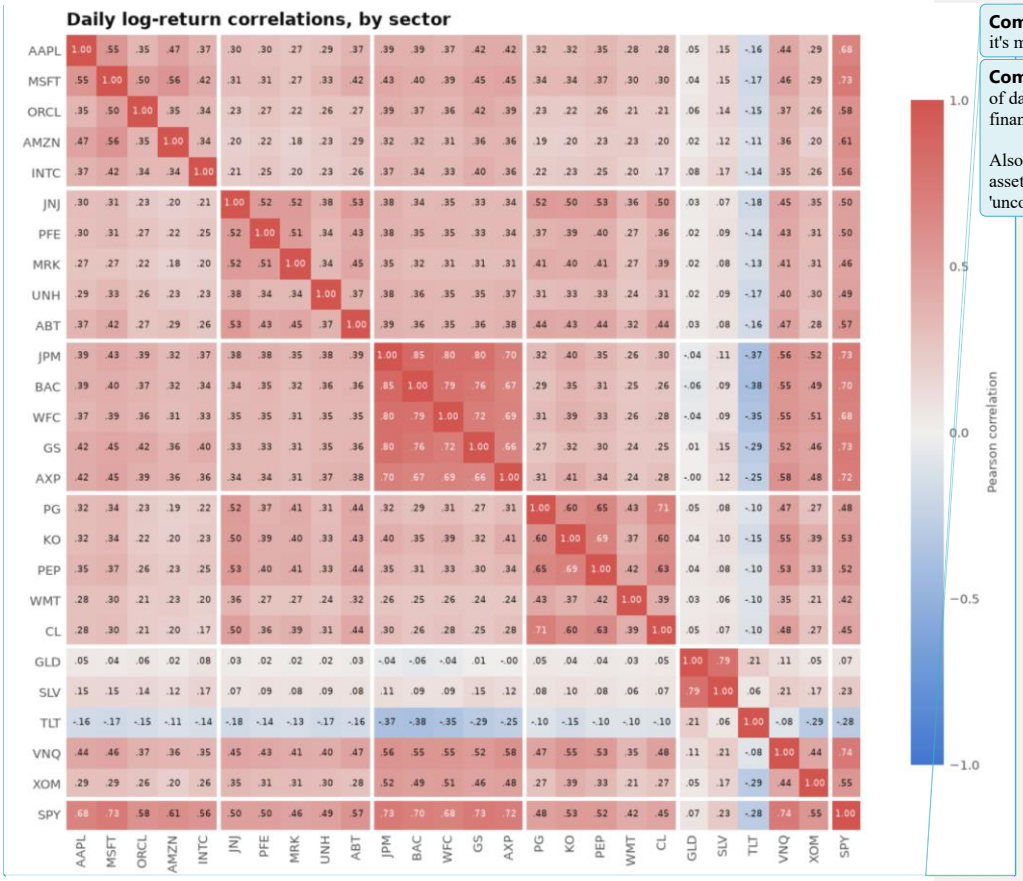
Appendix E — Reproduction instructions

[Provide the repository or code archive, commit/version identifier, environment and dependency versions, random seeds, configuration files and exact execution steps. Explain how to regenerate each main table and figure.]

Correlation Matrix (measured between 2010-01-01 and 2026-10-01)

Daily log-return correlations over the full-panel period (from the first date every asset has data) so every pair is measured on the same sample and the matrix is directly comparable cell to cell.

Assets are in the fixed sector order, with a gap between sector blocks so the block structure are clearly visible. SPY is shown in the last row and column as the external reference.



Comentado [CB6]: I have this in table format if you think it's more appropriate - we might want to add both

Comentado [CB7]: Notice strong intra-sector correlation of daily log returns observed within the assets from financials sector

Also notice how TLT is by far the most negatively correlated assets vs regular equities, and how GLD is the most 'uncorrelated with everything' - except silver.

Mean pairwise correlation:

- all 25 assets: 0.29
- within a sector : 0.46
- across sectors : 0.25

- Highest pair: ('JPM', 'BAC') at 0.85
- Lowest pair : ('BAC', 'TLT') at -0.38

Negatively correlated with how many of the other assets:

- TLT: 22
- GLD: 4
- WFC: 2
- JPM: 2

Sector	Mean within-sector corr.	Mean corr. with SPY
Technology	0.43	0.63
Healthcare	0.44	0.50
Financials	0.74	0.71
Consumer staples	0.55	0.48
Real assets & defensive	0.17	0.26

Comentado [CB8]: could paste it as a table as well...

The correlation heatmap shows daily returns interact following a structure: sector structure. Across all 25 the mean pairwise correlation is 0.29, but that average hides two different regimes. Inside a sector block, names move together: mean within-sector correlation is 0.46. The financials correlate at 0.74 among themselves (JPM:BAC 0.85, the highest pair in the matrix), making it the most correlated sector. Consumer staples at 0.55, technology and healthcare around 0.44. Across sectors the average falls to 0.25.

The diversifier assets are in one block (real assets and defensive)**.** TLT is negatively correlated with 22 of the other 24 assets, most strongly with the banks (about -0.35, since falling yields hurt bank margins and lift long bonds) and with SPY (-0.23). GLD is essentially uncorrelated with everything except SLV (0.79). VNQ and XOM, by contrast, behave like equities (0.71 and 0.52 with SPY), so the *Real assets & defensive* group is defensive mostly through TLT and GLD.

Correlation-aware strategies can use this structure for their benefit. With correlations being far from uniform (high within sectors, low across them, and negative for TLT), that unevenness is what separates the correlation-aware strategies (GMV, ERC, MDP, MDC) from the correlation-blind ones (EW, IV). When every pair is equally correlated, ERC and MDP reduce exactly to IV.
The contrast should therefore be largest when a subset's defensive asset composition includes a true diversifier (GLD or TLT). It should be smallest when there are no defensive assets or they are VNQ and XOM, that move similarly to equities.
A subset that draws two large banks hands the optimiser two assets at 0.80-0.85 correlation. Their difference being poorly estimated, unconstrained mean-variance weights may swing between them. Long-only constraints contain this, but it is the first place to look if MV or MSR turnover comes out high.